

Understanding Flow Matching from a mathematical view

Pipi Hu

Beijing Institute of Mathematical Sciences and Applications

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What I cannot create, I do not understand.

Mathematical motivation for Flow Matching

The fundamental challenge: Learn a velocity field $\mathbf{v}(\mathbf{x}, t)$ that transforms a simple base distribution $p_1(\mathbf{x})$ into the data distribution $p_0(\mathbf{x})$ through the continuity equation:

$$\partial_t p(\mathbf{x}, t) + \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0. \quad (1)$$

Key advantages over score-based methods:

- **No score estimation:** Avoids computing $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training
- **Analytic supervision:** Provides closed-form velocity targets via conditional paths
- **Deterministic sampling:** Enables ODE integration with large solver steps
- **Mathematical elegance:** Direct connection to optimal transport theory

Outline

- 1 Continuity equation and probability flow
- 2 Gaussian conditional paths and targets
- 3 Training and inference
- 4 Relation to diffusion and score models
- 5 Practical recipe
- 6 MeanFlow: Average Velocity Field
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The continuity equation: mathematical foundation

Theorem: Let $\mathbf{v}(\mathbf{x}, t)$ be a smooth velocity field. Then the density $p(\mathbf{x}, t)$ transported by the ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$ satisfies the continuity equation:

$$\partial_t p(\mathbf{x}, t) + \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0. \quad (2)$$

Key insight: This equation ensures probability conservation as particles flow according to the velocity field.

Physical meaning:

- $\partial_t p(\mathbf{x}, t)$: Rate of change of probability density
- $\nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t))$: Divergence of probability flux
- The equation states: "Change in density = Net flow in/out"

Proof of continuity equation: Step 1

Setup: For any test function $\phi(\mathbf{x})$, we have:

$$\frac{d}{dt} \int \phi(\mathbf{x}) p(\mathbf{x}, t) d\mathbf{x} = \int \phi(\mathbf{x}) \partial_t p(\mathbf{x}, t) d\mathbf{x} \quad (3)$$

Key insight: This is just the definition of the time derivative of the expectation.

Next step: We need to relate this to the particle motion governed by the ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$.

Proof of continuity equation: Step 2

Reynolds transport theorem: For a density $p(\mathbf{x}, t)$ transported by the velocity field $\mathbf{v}(\mathbf{x}, t)$, we have:

$$\frac{d}{dt} \int \phi(\mathbf{x}) p(\mathbf{x}, t) d\mathbf{x} = \int \phi(\mathbf{x}) \frac{\partial p(\mathbf{x}, t)}{\partial t} d\mathbf{x} + \int \phi(\mathbf{x}) \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) d\mathbf{x} \quad (4)$$

Key insight: The first term is the local change in density, the second term accounts for particles flowing in/out of the region.

Proof of continuity equation: Step 2 (continued)

Alternative approach - Direct calculation: Since particles follow $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$, the density evolution is:

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = -\nabla \cdot (p(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)) \quad (5)$$

Substituting back: From Step 1, we have:

$$\frac{d}{dt} \int \phi(\mathbf{x}) p(\mathbf{x}, t) d\mathbf{x} = \int \phi(\mathbf{x}) \frac{\partial p(\mathbf{x}, t)}{\partial t} d\mathbf{x} \quad (6)$$

$$= \int \phi(\mathbf{x}) [-\nabla \cdot (p(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t))] d\mathbf{x} \quad (7)$$

$$= - \int \phi(\mathbf{x}) \nabla \cdot (p(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)) d\mathbf{x} \quad (8)$$

Key insight: We use the fact that the density evolution equation is exactly the continuity equation!

Proof of continuity equation: Step 3

Integration by parts: Applying integration by parts to the right-hand side:

$$\int \nabla \phi(\mathbf{x}) \cdot \mathbf{v}(\mathbf{x}, t) p(\mathbf{x}, t) d\mathbf{x} = - \int \phi(\mathbf{x}) \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) d\mathbf{x} \quad (9)$$

Key insight: This converts the gradient of the test function to the divergence of the probability flux.

Proof of continuity equation: Conclusion

Combining results: From equations (3), (8), and (9):

$$\int \phi(\mathbf{x}) \partial_t p(\mathbf{x}, t) d\mathbf{x} = - \int \phi(\mathbf{x}) \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) d\mathbf{x} \quad (10)$$

Final step: Since this holds for any test function $\phi(\mathbf{x})$, we must have:

$$\boxed{\partial_t p(\mathbf{x}, t) + \nabla \cdot (p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0} \quad (11)$$

Conclusion: Any velocity field that transports particles via an ODE automatically satisfies the continuity equation!

Derivation: target velocity for Gaussian path

Given $\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$,

$$\frac{d}{dt} \mathbf{x}_t = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \boldsymbol{\epsilon}. \quad (12)$$

Condition on $\mathbf{x}_t = \mathbf{x}$ and $\mathbf{x}_0$. Since $\boldsymbol{\epsilon} = (\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0))/\sigma_t$,

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \mathbb{E}\left[\frac{d}{dt} \mathbf{x}_t \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0\right] \quad (13)$$

$$= \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0] \quad (14)$$

$$= \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \frac{\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t} \quad (15)$$

$$= \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0)). \quad (16)$$

Conditional probability paths: mathematical framework

Definition: Given a data distribution $p_{\text{data}}(\mathbf{x}_0)$ and base distribution $p_{\text{base}}(\mathbf{x})$, we define conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0)$ such that:

$$p_t(\mathbf{x}) = \int p_{\text{data}}(\mathbf{x}_0) q_t(\mathbf{x} \mid \mathbf{x}_0) d\mathbf{x}_0, \quad t \in [0, 1], \quad (17)$$

with boundary conditions $p_0 = p_{\text{data}}$ and $p_1 = p_{\text{base}}$.

Key insight: The conditional velocity field

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) := \mathbb{E}\left[\frac{d}{dt} \mathbf{x}_t \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0\right] \quad (18)$$

provides *analytic supervision* for learning the marginal velocity field $\mathbf{v}(\mathbf{x}, t)$.

Mathematical advantage: No need to estimate $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training!

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Gaussian conditional paths: mathematical derivation

Assumption: Gaussian conditional paths

$$q_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t(\mathbf{x}_0), \sigma_t^2 \mathbf{I}), \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}. \quad (19)$$

Mathematical derivation: Taking the time derivative of the path:

$$\frac{d}{dt} \mathbf{x}_t = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} = \frac{\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t}. \quad (20)$$

Conditional expectation: Since $\boldsymbol{\epsilon}$ is independent of $\mathbf{x}_t$ given $\mathbf{x}_0$, we have:

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \mathbb{E}\left[\frac{d}{dt} \mathbf{x}_t \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0\right] \quad (21)$$

$$= \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_t = \mathbf{x}, \mathbf{x}_0] \quad (22)$$

$$= \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t \frac{\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t} \quad (23)$$

Final result:

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0)).$$

Mathematical analysis of path choices

1. Linear bridge (optimal transport):

- $\mu_t(\mathbf{x}_0) = (1 - t) \mathbf{x}_0, \sigma_t = t \sigma_{\max}$
- **Mathematical property:** Minimizes Wasserstein-2 distance
- **Velocity:** $\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \frac{\sigma_{\max}}{t}(\mathbf{x} - \mathbf{x}_0)$

2. VP-like (DDPM-inspired):

- $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0, \sigma_t = \sqrt{1 - \alpha^2(t)}$
- **Mathematical property:** Preserves energy structure
- **Velocity:** $\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \dot{\alpha}(t) \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \alpha(t) \mathbf{x}_0)$

3. VE-like (SMLD-inspired):

- $\mu_t(\mathbf{x}_0) = \mathbf{x}_0, \sigma_t = \sigma(t)$ increasing
- **Mathematical property:** Preserves data structure
- **Velocity:** $\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \frac{\dot{\sigma}(t)}{\sigma(t)}(\mathbf{x} - \mathbf{x}_0)$

Key insight: All choices provide *closed-form* $\mathbf{u}_t$ via the boxed formula!

Derivation: VE/VP target forms

VE-like. $\mu_t(\mathbf{x}_0) = \mathbf{x}_0$, $\sigma_t = \sigma(t)$. Then

$$\mathbf{u}_t = 0 + \frac{\dot{\sigma}_t}{\sigma_t} (\mathbf{x} - \mathbf{x}_0). \quad (25)$$

VP-like. $\mu_t(\mathbf{x}_0) = \alpha(t) \mathbf{x}_0$, $\sigma_t = \sqrt{1 - \alpha^2(t)}$. Using $\mathbf{u}_t = \dot{\mu}_t + (\dot{\sigma}_t/\sigma_t)(\mathbf{x} - \mu_t)$, we get

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0) = \dot{\alpha} \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} - \frac{\dot{\sigma}_t}{\sigma_t} \alpha \mathbf{x}_0 \quad (26)$$

$$= \frac{\dot{\sigma}_t}{\sigma_t} \mathbf{x} + \left(\dot{\alpha} - \alpha \frac{\dot{\sigma}_t}{\sigma_t} \right) \mathbf{x}_0. \quad (27)$$

No explicit form for $\dot{\sigma}_t/\sigma_t$ is required to train; schedules determine it numerically.

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Flow Matching: mathematical training objective

Theoretical foundation: Given conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0)$, the optimal velocity field satisfies:

$$\mathbf{v}^*(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x}} [\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0)] \quad (28)$$

Training procedure:

- 1 Sample $t \sim \mathcal{U}[0, 1]$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_t \sim q_t(\cdot \mid \mathbf{x}_0)$
- 2 Compute analytic target $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)$ from chosen path
- 3 Minimize regression loss

Mathematical objective:

$$J_{\text{FM}} = \mathbb{E} [\| \mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0) \|^2]. \quad (29)$$

Key mathematical advantage: No score estimation needed; all targets are analytic!

Mathematical optimality of Flow Matching

Theorem: The population minimizer of the Flow Matching objective

$$J_{\text{FM}}(\theta) = \mathbb{E}[\|\mathbf{v}_{\theta}(\mathbf{x}_t, t) - \mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)\|^2] \quad (30)$$

is the posterior-averaged velocity field:

$$\mathbf{v}^*(\mathbf{x}, t) = \mathbb{E}[\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0) \mid \mathbf{x}_t = \mathbf{x}, t]. \quad (31)$$

Mathematical proof:

Proof.

Let $Y = \mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)$, $X = (\mathbf{x}_t, t)$. For any predictor $g(X)$,

$$\mathbb{E}[\|Y - g(X)\|^2] = \mathbb{E}[\|Y - \mathbb{E}[Y \mid X]\|^2] + \mathbb{E}[\|g(X) - \mathbb{E}[Y \mid X]\|^2] \quad (32)$$

$$+ 2\mathbb{E}[(Y - \mathbb{E}[Y \mid X]) \cdot (g(X) - \mathbb{E}[Y \mid X])] \quad (33)$$

The cross term vanishes by the tower property, and the second term is minimized when $g(X) = \mathbb{E}[Y \mid X]$.

Key insight: Flow Matching learns the *optimal* velocity field that respects the continuity

Mathematical foundation of inference

Theoretical basis: The learned velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ defines a deterministic ODE:

$$\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t), \quad t \in [0, 1] \quad (34)$$

Mathematical properties:

- **Existence:** Under Lipschitz conditions, solutions exist and are unique
- **Consistency:** The ODE preserves the continuity equation
- **Stability:** Deterministic integration avoids accumulation of stochastic errors

Generation process:

- 1 Initialize $\mathbf{x}(1) \sim p_1$ (e.g., $\mathcal{N}(0, \mathbf{I})$)
- 2 Integrate ODE backwards: $\frac{d}{dt} \mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t)$, $t : 1 \rightarrow 0$
- 3 Use standard ODE solvers (Euler, RK4, Dormand-Prince, etc.)

Mathematical advantages:

- Deterministic sampling (reproducible)
- Fast with large solver steps
- No stochastic correctors needed

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Mathematical connection to diffusion models

Theoretical link: For a forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$, the probability flow ODE is:

$$\frac{d}{dt} \mathbf{x} = f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t). \quad (35)$$

Key mathematical insight: When Flow Matching uses Gaussian paths that match the SDE's marginal distributions p_t , the learned velocity field $\mathbf{v}_\theta$ approximates the probability flow drift:

$$\mathbf{v}_\theta(\mathbf{x}, t) \approx f(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \quad (36)$$

Crucial advantage: Flow Matching achieves this approximation *without* ever computing the score $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training!

Mathematical derivation: probability flow ODE

Setup: Forward SDE $d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) dB_t$ with density $p(\mathbf{x}, t)$ solving the Fokker-Planck equation:

$$\partial_t p = -\nabla \cdot (fp) + \frac{1}{2} g^2 \Delta p. \quad (37)$$

Key insight: A deterministic ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$ transports densities via the continuity equation:

$$\partial_t p + \nabla \cdot (p\mathbf{v}) = 0. \quad (38)$$

Mathematical derivation: Equating the two equations:

$$-\nabla \cdot (p\mathbf{v}) = -\nabla \cdot (fp) + \frac{1}{2} g^2 \Delta p \quad (39)$$

$$\Rightarrow \mathbf{v} = f - \frac{1}{2} g^2 \frac{\nabla p}{p} = f - \frac{1}{2} g^2 \nabla \log p. \quad (40)$$

Result: The probability flow ODE drift is $\mathbf{v} = f - \frac{1}{2} g^2 \nabla \log p$.

Specializing VE/VP-style paths

- VE-style: $\mu_t(\mathbf{x}_0) = \mathbf{x}_0$. Target becomes $\mathbf{u}_t = \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \mathbf{x}_0)$.
- VP-style: $\mu_t = \alpha(t)\mathbf{x}_0$, $\sigma_t = \sqrt{1 - \alpha^2(t)}$. Target is $\mathbf{u}_t = \dot{\alpha} \mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \alpha\mathbf{x}_0)$.

Averaging over $\mathbf{x}_0$ under the path recovers the probability flow drift forms used by diffusion models.

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Complete training recipe

Step-by-step procedure:

- 1 **Sample data:** $\mathbf{x}_0 \sim p_{\text{data}}, t \sim \mathcal{U}[0, 1], \epsilon \sim \mathcal{N}(0, \mathbf{I})$
- 2 **Forward pass:** $\mathbf{x}_t = \mu_t(\mathbf{x}_0) + \sigma_t \epsilon$
- 3 **Compute target:** $\mathbf{u}_t(\mathbf{x}_t | \mathbf{x}_0) = \dot{\mu}_t(\mathbf{x}_0) + (\dot{\sigma}_t / \sigma_t)(\mathbf{x}_t - \mu_t(\mathbf{x}_0))$
- 4 **Regression loss:** $\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}_t(\mathbf{x}_t | \mathbf{x}_0)\|^2$

Implementation notes:

- Use standard optimizers (Adam, AdamW)
- Typical learning rates: 10^{-4} to 10^{-3}
- Training time similar to diffusion models

Complete inference recipe

Generation procedure:

- ① **Initialize:** $\mathbf{x}(1) \sim p_1$ (e.g., $\mathcal{N}(0, \mathbf{I})$)
- ② **ODE integration:** $\dot{\mathbf{x}} = \mathbf{v}_\theta(\mathbf{x}, t)$ from $t = 1$ to $t = 0$
- ③ **Optional:** Add predictor-corrector steps for higher quality

Solver recommendations:

- **Fast:** Euler method (1-10 steps)
- **Accurate:** RK4, Dormand-Prince (10-50 steps)
- **Adaptive:** Use error control for automatic step sizing

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- Goal: map a simple prior (e.g., $\epsilon \sim \mathcal{N}(0, I)$) to the data distribution p_{data} .
- Diffusion / Flow Matching usually require *multi-step* ODE/SDE integration during sampling.
- Desire: **single-step (1-NFE)** generation while retaining high fidelity and stability.
- Key idea in MeanFlow: learn a *ground-truth average velocity field* that summarizes an entire time interval.

Rectified / Flow Matching setup

Given schedules a_t, b_t and time $t \in [0, 1]$:

$$\mathbf{z}_t = a_t \mathbf{x} + b_t \epsilon, \quad (41)$$

$$\mathbf{v}_t \equiv \frac{d\mathbf{z}_t}{dt} = a'_t \mathbf{x} + b'_t \epsilon. \quad (42)$$

Marginal velocity (ground-truth field for FM):

$$\mathbf{v}(\mathbf{z}_t, t) = \mathbb{E}_{p_t(\mathbf{v}_t|\mathbf{z}_t)}[\mathbf{v}_t]. \quad (43)$$

Training (conditional FM):

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, \mathbf{x}, \epsilon} \left\| v_\theta(\mathbf{z}_t, t) - \mathbf{v}_t \right\|_2^2, \quad \text{with } \mathbf{z}_t = a_t \mathbf{x} + b_t \epsilon. \quad (44)$$

Sampling in Flow Matching

The probability-flow ODE:

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}(\mathbf{z}_t, t), \quad (\text{start from } \mathbf{z}_1 = \epsilon). \quad (45)$$

With a discrete solver (Euler): $\mathbf{z}_{t_{i+1}} \approx \mathbf{z}_{t_i} + (t_{i+1} - t_i) \mathbf{v}(\mathbf{z}_{t_i}, t_i)$.

Issue

Coarse discretization along *curved* trajectories leads to significant numerical error $\Rightarrow$ many steps are needed.

Definition: Average Velocity

For any pair $r < t$ and point $\mathbf{z}_t$ along the flow path:

$$\bar{\mathbf{u}}(\mathbf{z}_t, r, t) := \frac{1}{t-r} \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau) d\tau. \quad (46)$$

- $\bar{\mathbf{u}}$ is an **underlying field** induced by $\mathbf{v}$ (no networks yet).
- As $r \rightarrow t$, $\bar{\mathbf{u}}(\mathbf{z}_t, r, t) \rightarrow \mathbf{v}(\mathbf{z}_t, t)$.
- Composition over intervals holds: one big step equals two smaller consecutive steps (additivity of integrals).

MeanFlow Identity (core relation)

Starting from Eq. (46):

$$(t - r) \bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau) d\tau. \quad (47)$$

Differentiate w.r.t. t (treating r as constant) and use the fundamental theorem of calculus:

$$\boxed{\bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \mathbf{v}(\mathbf{z}_t, t) - (t - r) \frac{d}{dt} \bar{\mathbf{u}}(\mathbf{z}_t, r, t)} \quad (48)$$

where the total derivative expands to

$$\frac{d}{dt} \bar{\mathbf{u}}(\mathbf{z}_t, r, t) = \underbrace{\mathbf{v}(\mathbf{z}_t, t)}_{\frac{d\mathbf{z}_t}{dt}} \cdot \partial_{\mathbf{z}} \bar{\mathbf{u}}(\mathbf{z}_t, r, t) + \partial_t \bar{\mathbf{u}}(\mathbf{z}_t, r, t). \quad (49)$$

Interpretation: Eq. (48) links instantaneous and average velocities *without* introducing networks.

Trainable objective

Parameterize a network $\bar{\mathbf{u}}_\theta(\mathbf{z}, r, t)$. Use the MeanFlow identity to form a target (replace $\mathbf{v}$ by sample-conditional $\mathbf{v}_t$):

$$\text{Target: } \bar{\mathbf{u}}_{\text{tgt}} = \mathbf{v}_t - (t - r) \underbrace{(\mathbf{v}_t \cdot \partial_{\mathbf{z}} \bar{\mathbf{u}}_\theta + \partial_t \bar{\mathbf{u}}_\theta)}_{\text{JVP}}. \quad (50)$$

$$\mathcal{L}(\theta) = \mathbb{E} \left\| \bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \text{sg}(\bar{\mathbf{u}}_{\text{tgt}}) \right\|_2^2. \quad (51)$$

- $\text{sg}(\cdot)$: stop-gradient; avoids higher-order differentiation.
- JVP (Jacobian-vector product) computed efficiently via autodiff.
- If $r = t$, the correction term vanishes $\Rightarrow$ reduces to standard Flow Matching.

Pseudocode (training)

MeanFlow training algorithm:

- 1 Sample $t, r \sim \text{LogitNormal}$; set $t \leftarrow \max(r, t)$, $r \leftarrow \min(r, t)$
- 2 Sample $\mathbf{x} \sim p_{\text{data}}$, $\epsilon \sim \mathcal{N}(0, I)$; set $\mathbf{z}_t = (1 - t)\mathbf{x} + t\epsilon$, $\mathbf{v}_t = \epsilon - \mathbf{x}$
- 3 Compute $(\bar{\mathbf{u}}_\theta, \frac{d}{dt}\bar{\mathbf{u}}_\theta) = \text{JVP}(\bar{\mathbf{u}}_\theta(\cdot, r, t), (\mathbf{v}_t, 0, 1))$
- 4 Set $\bar{\mathbf{u}}_{\text{tgt}} \leftarrow \mathbf{v}_t - (t - r) \frac{d}{dt}\bar{\mathbf{u}}_\theta$
- 5 Minimize $\|\bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \text{sg}(\bar{\mathbf{u}}_{\text{tgt}})\|_2^2$ (optionally with adaptive weights)

Sampling (1 step or few steps)

Replace the time integral by the learned average velocity:

$$\mathbf{z}_r = \mathbf{z}_t - (t - r) \bar{\mathbf{u}}_\theta(\mathbf{z}_t, r, t). \quad (52)$$

1-NFE sampling: take $(r, t) = (0, 1)$, $\mathbf{z}_1 = \epsilon$,

$$\boxed{\mathbf{x} \equiv \mathbf{z}_0 = \epsilon - \bar{\mathbf{u}}_\theta(\epsilon, 0, 1)} . \quad (53)$$

- Few-step variants are immediate by chaining intervals.

CFG inside the field (still 1-NFE)

Define a CFG velocity field

$$\mathbf{v}_{\text{cfg}}(\mathbf{z}_t, t \mid c) = \omega \mathbf{v}(\mathbf{z}_t, t \mid c) + (1 - \omega) \mathbf{v}(\mathbf{z}_t, t). \quad (54)$$

Key insight: CFG can be applied directly to the average velocity field:

$$\bar{\mathbf{u}}_{\text{cfg}}(\mathbf{z}_t, r, t \mid c) = \omega \bar{\mathbf{u}}(\mathbf{z}_t, r, t \mid c) + (1 - \omega) \bar{\mathbf{u}}(\mathbf{z}_t, r, t). \quad (55)$$

This maintains the 1-NFE property while enabling conditional generation.

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Key references and resources

Core papers:

- Lipman et al., *Flow Matching: A unifying perspective of score-based training and generative modeling* (Guide and Code, 2024)
- Song et al., *Score-based generative modeling through stochastic differential equations* (2021)
- Ho et al., *Denoising Diffusion Probabilistic Models* (2020)

Implementation resources:

- Flow Matching Guide and Code (GitHub)
- Open source libraries: diffusers, torchdyn
- Tutorial notebooks and examples

Welcome inputs

Any comments?

Welcome your inputs!